Question 1 Calculator Allowed

Markscheme

Difficulty: **Easy**

Which of the following correctly states the conclusion drawn from the Rutherford-Geiger-Marsden experiment?

- A. The nucleus of an atom contains protons and neutrons.
- B. Protons and electrons have equal but opposite charges.
- C. The nucleus contains nearly the entire mass of the atom.
- D. Gold atoms absorb alpha particles.

Question 2 Calculator Allowed

Markscheme

Difficulty: **Easy**

Three experiments are listed.

- I. Photoelectric effect
  II. Davisson-Germer experiment
  III. Young's double-slit experiment

Which of the experiments show the existence of the wave behavior of matter?

- A. II only
- B. II and III only
- C. I and III only
- D. I, II and III only

Question 3 Calculator Allowed

Markscheme

Difficulty: **Easy**

Which of the following physical quantities has the same SI unit as the radioactive decay constant?

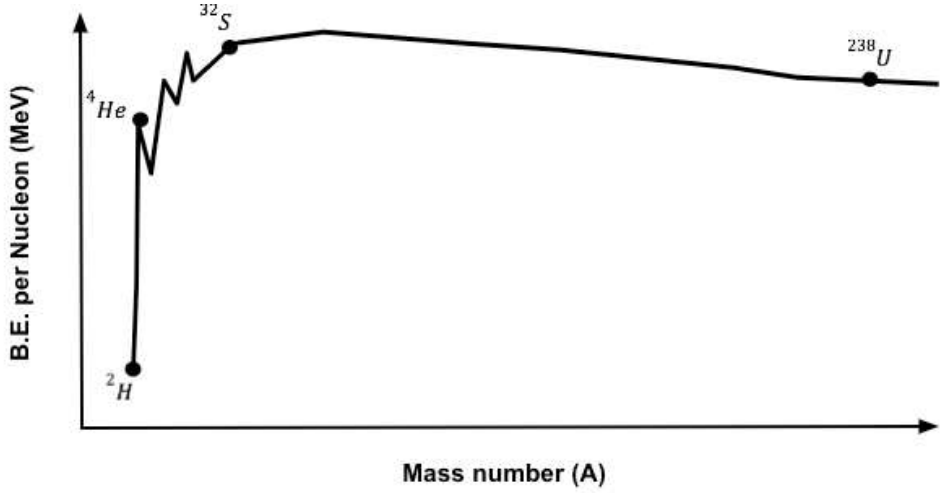
- A. Frequency
- B. Time period
- C. Wavelength
- D. Wave speed

Question 4 Calculator Allowed

Markscheme

Difficulty: **Easy**

The diagram below shows the graph of binding energy per nucleon vs. the mass (nucleon) number of some elements.



Which is the most stable nucleus?

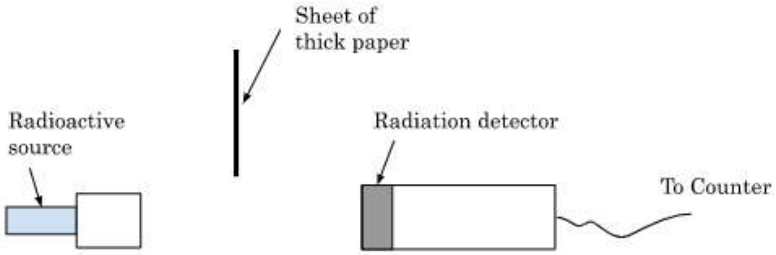
- A. $^{238}\mathrm{U}$
- B. $^{2}\mathrm{H}$
- C. $^{4}\mathrm{He}$
- D. $^{32}\mathrm{S}$

Question 5 Calculator Allowed

Markscheme

Difficulty: **Easy**

A student uses the arrangement shown below to ascertain the type of radiation emitted by a given sample of radioactive material.



When the sheet of paper is in the position shown, a count rate of 720 Bq is recorded. When she lowers the thick sheet of paper so as to obstruct the direct path between the source and the detector, the count rate drops to 18 Bq. Which of the following is the most likely explanation for the data collected?

- A. Only alpha particles are emitted. The remaining activity is from background radiation.
- B. An equal number of alpha and beta particles are emitted.
- C. Only beta particles are emitted.
- D. There is not enough data to draw a reasonable conclusion.

Question 6 Calculator Allowed

Markscheme

Difficulty: **Easy**

A nuclear reactor converts nuclear energy into electrical energy. Which of the following reactions best describes the primary nuclear reaction in a nuclear reactor?

- A. $^{234}_{90}\mathrm{Th} \rightarrow {}^{0}_{-1}\beta + {}^{234}_{91}\mathrm{Pa} + {}^{0}_{0}\overline{\nu}$
- B. $^{232}_{92}\mathrm{U} \rightarrow {}^{228}_{90}\mathrm{Th} + {}^{4}_{2}\alpha$
- C. $^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{90}_{36}\mathrm{Kr} + {}^{143}_{56}\mathrm{Ba} + 3\,{}^{1}_{0}n$
- D. $^{14}_{6}\mathrm{C} \rightarrow {}^{7}_{3}\mathrm{Li} + {}^{7}_{3}\mathrm{Li}$

Question 7 Calculator Allowed

Markscheme

Difficulty: **Easy**

Which statement describes a possible nuclear fusion reaction?

- A. Hydrogen nuclei join together to form a helium nucleus.
- B. A helium nucleus joins with a hydrogen nucleus to form an alpha particle.
- C. Uranium nuclei split and produce high energy neutrons causing a chain reaction.
- D. Helium nuclei join to form hydrogen nuclei.

Question 8 Calculator Allowed

Markscheme

Difficulty: **Easy**

Which of the following is correct for a stable star?

- A. Inwards force due to radiation pressure equals the outwards force due to gravitational attraction
- B. Inwards force due to radiation pressure is greater than the outwards force due to gravitational attraction
- C. Inwards force due to gravitational attraction equals the outwards force due to radiation pressure
- D. Inwards force due to gravitational attraction is greater than the outwards force due to radiation pressure

Question 9 Calculator Allowed

Markscheme

Difficulty: **Easy**

Which of the following is not true about Bohr's model of the atom.

- A. It explains the stability of electrons within their energy levels.
- B. It correctly calculates the energies of electrons in all atoms.
- C. It explains the quantisation of the energy of hydrogen atoms.
- D. It assumes that the angular momentum of electrons is quantised.

Question 10 Calculator Allowed

Markscheme

Difficulty: **Easy**

An electron in a hydrogen atom is excited from the ground state of $n = 1$ to the $n = 4$ level. What is the total number of frequencies that can be observed from the emission spectrum as the electron returns to ground state?

- A. 12
- B. 6
- C. 4
- D. 3

Question 11 Calculator Allowed

Markscheme

Difficulty: **Easy**

In a vacuum chamber, electromagnetic radiation of frequencies $f_A < f_B$ just manage to produce a non-zero photocurrent from metals A and B respectively. When illuminated by an electromagnetic radiation of frequency $f > f_B$, the stopping voltage for the photocurrents from plates A and B are recorded to be V_A and V_B respectively.
Which of the following is true about V_A and V_B?

- A. $V_A < V_B$
- B. $V_A > V_B$
- C. $V_A = V_B$
- D. Information provided is insufficient to compare V_A and V_B

Question 12 Calculator Allowed

Markscheme

Difficulty: **Easy**

Two lasers, A and B emit light of the same wavelength. If the intensity of A is greater than that of B, which of the following statements would be the correct interpretations of this situation using the wave and particle models of light.

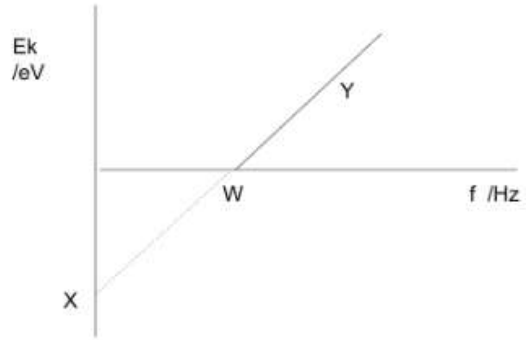
	Wave model	Particle model
A.	The amplitude of LASER A is greater than that of LASER B.	LASER A emits more photons per second than LASER B.
B.	The amplitude of LASER B is greater than that of LASER A.	LASER B emits more photons per second than LASER A.
C.	The amplitude of LASER A is greater than that of LASER B.	LASER B emits more photons per second than LASER A.
D.	The amplitude of LASER B is greater than that of LASER A.	LASER A emits more photons per second than LASER B.

Question 13 Calculator Allowed

Markscheme

Difficulty: **Easy**

The graph below shows the graph of the maximum kinetic energy (E_k)of the emitted electrons against incident frequency (f) plotted by a student investigating the work function for a particular metal.



Which of the following can be used by the student to get an approximate value for the work function of the metal in question?

- A. Intercept X
- B. Intercept W
- C. Gradient at Y
- D. $\frac{\text{Speed of Light}}{\text{Intercept W}}$

Question 14 Calculator Allowed

Markscheme

Difficulty: **Easy**

The binding energy of ${}^4_2He$ is 28 MeV. What is its binding energy per nucleon?

- A. 45×10^{-13} J
- B. 22×10^{-13} J
- C. 11×10^{-13} J
- D. 7.5×10^{-13} J

Question 15 Calculator Allowed

Markscheme

Difficulty: **Easy**

In the absence of the strong nuclear force, a large nucleus would be even more unstable than a small nucleus. Which one of the following is the best explanation?

- A. A nucleus with fewer protons will more likely stay together due to the gravitational attraction between the nucleons.
- B. The more protons in the nucleus, the more repulsive force there would be to make it unstable.
- C. A nucleus with fewer neutrons will more likely stay together due to the gravitational attraction between the nucleons.
- D. The more neutrons there are in the nucleus, the larger the separation of the protons which weakens the electrostatic repulsion between the protons.

Question 16 Calculator Allowed

Markscheme

Difficulty: **Easy**

An element X is bombarded with alpha particles and a new element Y is formed with a neutron emitted. The following equation can be written for the nuclear reaction.

$$ {}^4_2\mathrm{He} + {}^{\square}_{A}\mathrm{X} \rightarrow {}^{\square}_{B}\mathrm{Y} + \mathrm{n} $$

The atomic numbers of X and Y are A and B respectively. Which of the following is correct for A and B?

- A. $A - 1 = B$
- B. $A + 1 = B$
- C. $A - 2 = B$
- D. $A + 2 = B$

Question 17 Calculator Allowed

Markscheme

Difficulty: **Easy**

Consider the following statements about nuclear binding energy:
The nuclear binding energy is:

- I. proportional to the mass defect from nucleus formation.
- II. the energy released when a nucleus is formed from its nucleons.
- III. the energy required to remove a proton from the nucleus.

Which of the statements are correct?

- A. I only
- B. I and II only

- C. II only
- D. I, II and III

Question 18 Calculator Allowed

Markscheme

Difficulty: **Easy**

A nuclide of a radioactive element Q can be represented as ${}^{234}_{90}\mathrm{Q}$. The nuclide undergoes two negative beta decays and one alpha decay. Which of the following configurations correctly represents the daughter nuclide R?

- A. ${}^{234}_{92}\mathrm{R}$
- B. ${}^{232}_{92}\mathrm{R}$
- C. ${}^{230}_{90}\mathrm{R}$
- D. ${}^{230}_{92}\mathrm{R}$

Question 19 Calculator Allowed

Markscheme

Difficulty: **Easy**

[Maximum mark: 7]

1.
A star X is burning carbon in its core. The reaction is given by:
   - 2. ${}^{12}_{6}\mathrm{C} + {}^{12}_{6}\mathrm{C} \rightarrow {}^{24}_{12}Mg$

The following data are available for the reaction.
   - 3.
     Rest mass of the ${}^{12}_{6}\mathrm{C} = 12.0111507\,u$
   - 4.
     Rest mass of the ${}^{24}_{12}Mg = 23.9909630\,u$

Determine, in MeV, the energy released in the reaction. Be sure to express your answer to the correct number of significant figures [3]

5.
The star Y is a red supergiant. State the next steps in the evolutionary sequence of the star Y.

6.
Outline why the sun is not expected to undergo carbon fusion. [2]

Question 20 Calculator Allowed

Markscheme

Difficulty: **Easy**

Which of the following is correct about a red giant star?

- I. The orbits of the inner planets of the solar system and could fit into the volume of it
  II. Cooler than the Sun
  III. Have higher luminosity than the Sun

- A. I only
- B. II only
- C. I and II only
- D. I and III only

Question 21 Calculator Allowed

Markscheme

Difficulty: **Easy**

A student investigates the scattering of alpha particles. Initially, she uses a gold foil (Z = 79) and next she repeats the experiment using an aluminum (Z = 13) foil of the same thickness. The scattered alpha particles are observed on a detector that fluoresces.
Which of the following will be true as she switches from the gold foil to the aluminum foil?

- I. The alpha particles deflect more from the gold nucleus than the aluminum.
  II. The alpha particles approach the gold nucleus closer than the aluminum nucleus.
  III. There is no change in the observed scattering pattern.

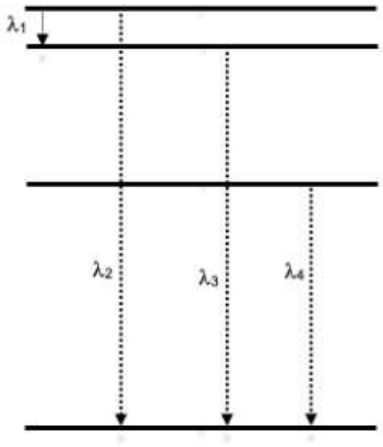
- A. I only
- B. II only
- C. III only
- D. II and III only

Question 22 Calculator Allowed

Markscheme

Difficulty: **Easy**

The diagram shows atomic energy levels of an element, and wavelengths are labeled corresponding to emission for electron transition between certain energy levels.



Which of the following shows these wavelengths in decreasing order?

- A. $\lambda_1, \lambda_2, \lambda_3, \lambda_4$
- B. $\lambda_4, \lambda_3, \lambda_2, \lambda_1$
- C. $\lambda_1, \lambda_4, \lambda_3, \lambda_2$
- D. $\lambda_2, \lambda_3, \lambda_4, \lambda_1$

Question 23 Calculator Allowed

Markscheme

Difficulty: **Easy**

[Maximum mark: 7]

1. With reference to energy conversion, distinguish between photovoltaic cells and solar heating panels. [2]
2.

On a sunny day, a house is powered by a solar panel. In order to run home appliances, the incident power of sunlight on the solar panel must be $10\,kW$.

1.
The average surface temperature of the Sun is $5850\,K$. Show that the peak wavelength of the emission of the sun is around $500\,nm$. [2]

2.
It can be assumed that 7.5% of the energy from incident sunlight consists of peak wavelength photons that are converted to electrical energy by the panel. Calculate the number of $500\,nm$ photons that must strike the solar panel per second in order to meet the power requirement of the appliances. [3]

Question 24 Calculator Allowed

Markscheme

Difficulty: **Easy**

In the design of a nuclear fission reactor, which of the following statements are true?

- I. Control rods can increase the rate of fission when inserted into the reactor
  II. The moderator blocks neutrons to ensure the chain reaction is controlled
  III. Enrichment of uranium increases the probability of a chain reaction taking place

- A. I and III only
- B. II and III only
- C. I and II only
- D. III only

Question 25 Calculator Allowed

Markscheme

Difficulty: **Easy**

The nucleon numbers of elements X and Y are A and $8A$ respectively. Their radii are R_X and R_Y, and their densities are ρ_X and ρ_Y.

Which of the following shows the correct values of the ratios $\frac{R_Y}{R_X}$ and $\frac{\rho_Y}{\rho_X}$?

	$\frac{R_Y}{R_X}$	$\frac{\rho_Y}{\rho_X}$
A.	2	1
B.	8	2
C.	2	2
D.	8	1

Question 26 Calculator Allowed

Markscheme

Difficulty: **Easy**

Which value below gives the closest approximation for the diameter of a uranium-238 nucleus?

- A. $10^{-16}\,m$
- B. $10^{-14}\,m$
- C. $10^{-12}\,m$

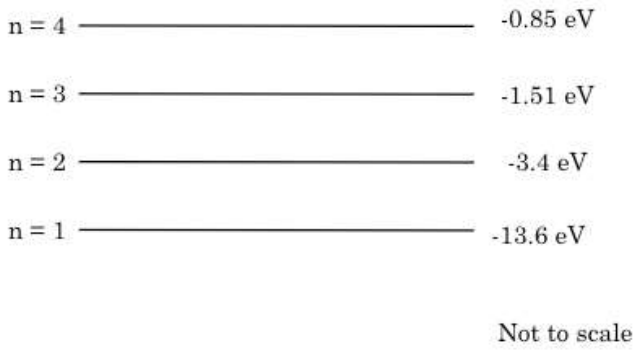
- D. $10^{-10}\,m$

Question 27 Calculator Allowed

Markscheme

Difficulty: **Easy**

The electron in the ground state of a hydrogen atom absorbs a certain amount of energy. The diagram shows some of the energy levels of the hydrogen atom. At which state would the electron be if it absorbs 12.09 eV of energy?



- A. At ground state.
- B. First excited state.
- C. Second excited state.
- D. Third excited state.

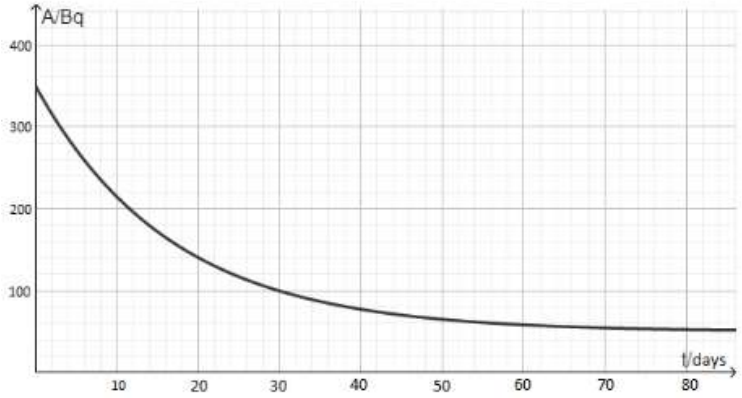
Question 28 Calculator Allowed

Markscheme

Difficulty: **Easy**

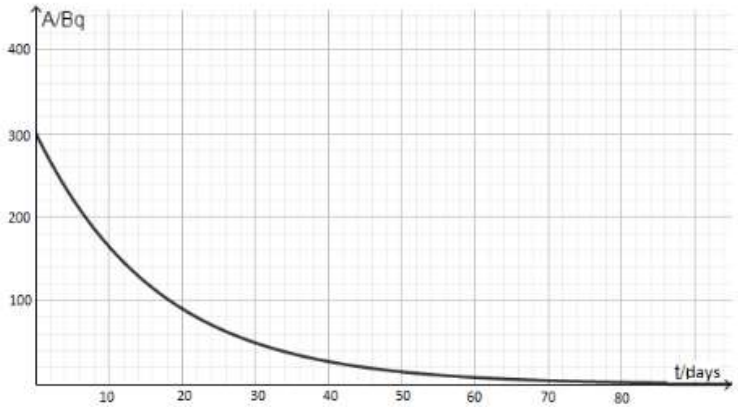
[Maximum mark: 6]

The graph shows the variation of the measured activity of an isotope sample with time.



[Source: Created with GeoGebra, - https://www.geogebra.org/graphing?lang=en]

1. State what is meant by activity. [1]
2. Explain why the activity graph doesn't approach zero as time increases. [2]
3. The activity graph is corrected so as to reach zero as time increases as shown below.



[Source: Created with GeoGebra, - https://www.geogebra.org/graphing?lang=en]

1.
Determine the half-life of the sample. [1]

2.
The initial mass of this sample was 248 g. Determine the time that would elapse before 31 g of the sample remained. [2]

Question 29 Calculator Allowed

Markscheme

Difficulty: **Easy**

A 1.20 kg sample of a radioactive isotope decaying via beta decay with a half-life of 30 minutes is observed in a closed container in a laboratory. The decay results in a daughter nucleus that is also a solid. What is the approximate total mass measured after 90 minutes?

- A. 0.00 kg
- B. 0.15 kg
- C. 0.60 kg
- D. 1.20 kg

Question 30 Calculator Allowed

Markscheme

Difficulty: **Easy**

The activity of a 100% pure radioactive sample is $10^6\ Bq$. The activity of the sample after 20% of nuclei have decayed is:

- A. $0.2 \times 10^6\ Bq$
- B. $0.4 \times 10^6\ Bq$
- C. $0.6 \times 10^6\ Bq$
- D. $0.8 \times 10^6\ Bq$

Question 31 Calculator Allowed

Markscheme

Difficulty: **Easy**

During experiments observing beta decays, it was seen that the energy spectrum of the beta particles is continuous and momentum conservation was violated in beta decays. This provided evidence for the existence of:

- A. Nuclear energy levels
- B. Atomic energy levels
- C. Neutrinos

- D. Positrons

Question 32 Calculator Allowed

Markscheme

Difficulty: **Easy**

Which of the following radioactive decays can cause the production of a daughter nucleus to be a different element than the parent nucleus?

- I. α decay
  II. β decay
  III. γ decay

- A. I only
- B. II only
- C. I and II only
- D. I, II, and III

Question 33 Calculator Allowed

Markscheme

Difficulty: **Easy**

In a nuclear reactor, what is the name given to the material used to slow down neutrons in order to cause successful fissions?

- A. Control rod
- B. Moderator
- C. Heat exchanger
- D. Chain reaction

Question 34 Calculator Allowed

Markscheme

Difficulty: **Easy**

Which of the following is incorrectly matches a component of a nuclear power plant with its function?

	Component	Function
A.	Heat Exchanger	Prevents overheating
B.	Moderator	Slows down electrons
C.	Control rods	Absorbs neutrons
D.	Shielding	Prevents any leakage of radiation

Question 35 Calculator Allowed

Markscheme

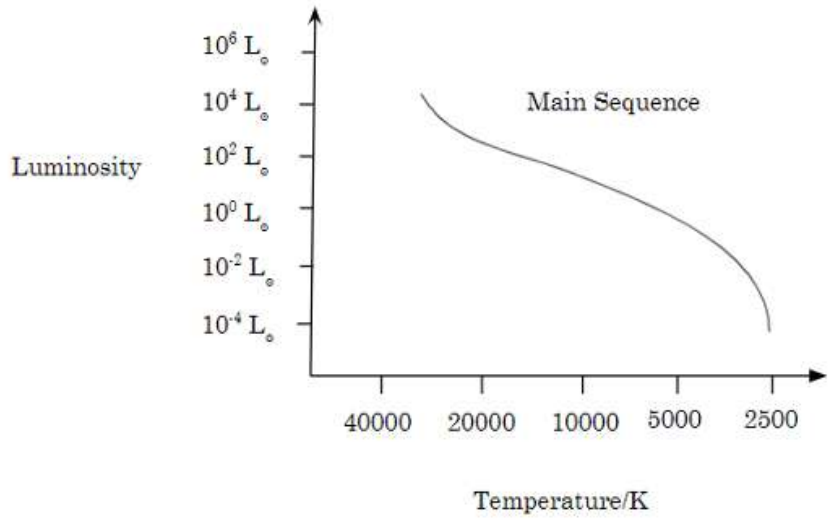
Difficulty: **Easy**

[Maximum mark: 14]

1. VV Cephei is a binary star system composed of two stars referred to as VV Cephei A and VV Cephei B.
   1. The measured wavelength of the peak spectral line of VV Cephei A is $760\,nm$. Show that the surface temperature of VV Cephei A is approximately $3\,800\,K$. [2]
   2.

The surface temperature of the Sun is $T_{\odot} = 5800K$. The luminosity of the VV Cephei A is 2.7×10^5 times the Sun's luminosity $L_{\odot}$. Determine the radius of VV Cephei A in terms of the Sun's radius $R_{\odot}$. [2]

3.
A Herzsprung-Russel (HR) diagram is shown.



Plot the current position of VV Cephei A on the HR diagram and identify the classification of VV Cephei A. [2]

1.
The mass of VV Cephei B is about 11 times the mass of the Sun and it is not on the main sequence.

1. In terms of forces, explain the condition that causes a star to first leave the main sequence. [2]
2. Explain why VV Cephei B is likely to remain on the main sequence for a shorter time than the Sun. [2]
3. The parallax angle for VV Cephei B is 0.31 arcseconds. Calculate, in light years, the distance between the earth and VV Cephei B. [2]
4. State why the parallax method is not suitable for measuring the distance of very distant stars. [2]

Question 36 Calculator Allowed

Markscheme

Difficulty: **Easy**

A main sequence star has a mass greater than 4 times the mass of the Sun. Which of the following is correct about the possible the fate of the star?

- I. White dwarf
  II. Neutron star
  III. Black hole

- A. I only
- B. II only
- C. I and II only
- D. II and III only

Question 37 Calculator Allowed

Markscheme

Difficulty: **Easy**

A star with a mass of twice the mass of the sun is a main sequence star. The time the sun is expected to spend in the main sequence is T. Which of the following is

correct for the time the star is expected to spend in the main sequence and the next stage in its evolution?

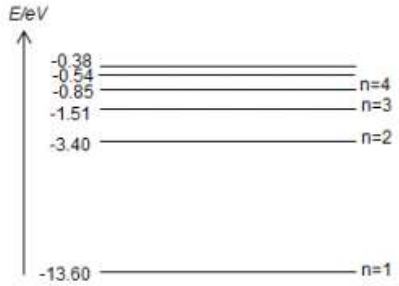
	Time	Stage
A.	More than T	Red giant
B.	More than T	Supergiant
C.	Less than T	Red giant
D.	Less than T	Supergiant

Question 38 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 10]
The diagram shows six energy levels of hydrogen.



1. Describe how an emission spectrum is an evidence of discrete energy levels in an atom. [2]
2. 1. On the energy level diagram, draw an arrow to represent the transition that emits the shortest wavelength. [2]
   2. Calculate the shortest emitted wavelength. [3]
3. Emitted red light has a frequency estimation of 460 THz. Determine which energy level transition emits this red light. [3]

Question 39 Calculator Allowed

Markscheme

Difficulty: **Medium**

Alpha particles of kinetic energy K are fired at nuclei of proton number Z.
The number of particles scattered at different angles are in agreement with the prediction of Rutherford's scattering model.
Which of the following changes to the values of K and Z will result in a significant deviation from Rutherford's model?

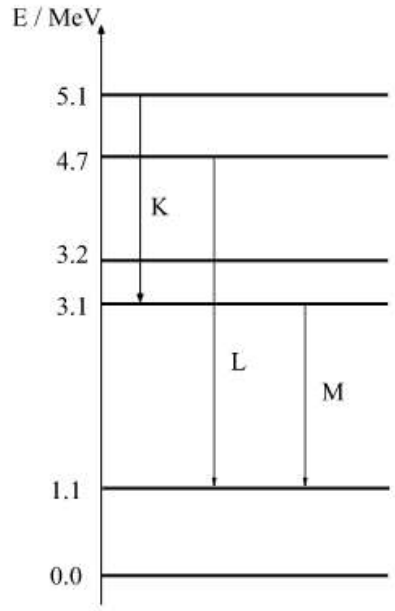
	Value of K	Value of Z
A.	Decrease	Increase
B.	Decrease	Decrease
C.	Increase	Increase
D.	Increase	Decrease

Question 40 Calculator Allowed

Markscheme

Difficulty: **Medium**

The nuclear energy levels of a nucleus are shown below. K, L, and M are transitions between energy levels.



Which statements are correct about photons emitted during transitions K, L, and M?

- I. The wavelength of the photon emitted by transition K is greater than the wavelength of the photon emitted by transition L.
  II. The frequency of the photon emitted by transition K is equal to the frequency of the photon emitted by transition M.
  III. The wavelength of the photon emitted by transition L is greater than the wavelength of the photon emitted by transition M.

- A. I only
- B. I and II only
- C. II and III only
- D. I, II and III

Question 41 Calculator Allowed

Markscheme

Difficulty: **Medium**

Light from a sodium lamp of wavelength is directed towards a metal surface and electrons until a stopping potential of V_s is applied. The wavelength of the incident light is halved. Which of the following corresponds to the new value of the required stopping potential?

- A. $\frac{V_s}{2}$
- B. $\frac{V_s}{2} - \frac{\phi}{e}$
- C. $2V_s$
- D. $2V_s + \frac{\phi}{e}$

Question 42 Calculator Allowed

Markscheme

Difficulty: **Medium**

No electrons are emitted when metal plate X is irradiated by an electromagnetic radiation of wavelength λ, and intensity I. Which of the following could result in emission of photoelectrons?

- I. Decreasing the wavelength λ of incident light.
  II. Increasing the intensity of electromagnetic radiation at the same wavelength.
  III. Replacing metal X, with a metal of smaller threshold frequency.

- A. I and II only
- B. I and III only
- C. II and III only
- D. I, II and III

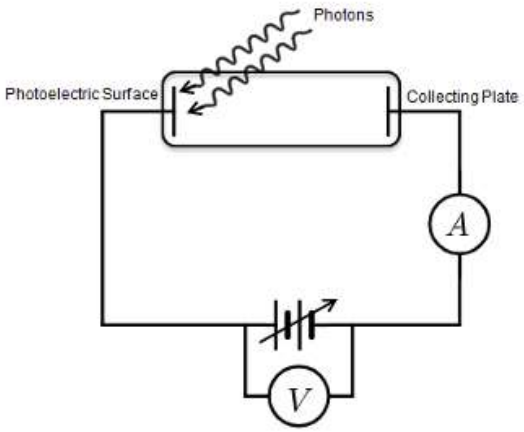
Question 43 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 12]

1.
Light falls on a photoelectric surface with a threshold frequency of $5.2 \times 10^{14}\ Hz$.



[Source: Created with GeoGebra by Tom Walsh- https://www.geogebra.org/m/JbAKa82Y]

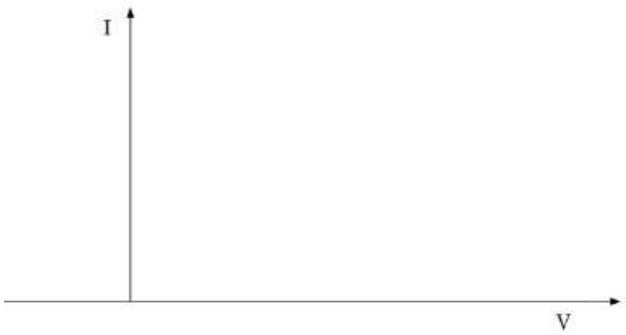
State what is meant by photoelectric emission. [1]

2.
1.
Calculate the work function of the photoelectric surface in electronvolts. [2]

2.
The ammeter gives a reading when a violet light is used, while it reads zero when a red light with the same intensity is used. Explain this observation. [3]

3.
The frequency of violet photons is $7.5 \times 10^{14}\ Hz$. Calculate the maximum speed of the emitted electrons. [3]

3.
On the axes below, sketch the variation of the photoelectric current with the potential difference for an incident frequency above the threshold frequency. Also label the stopping voltage V_s. [3]



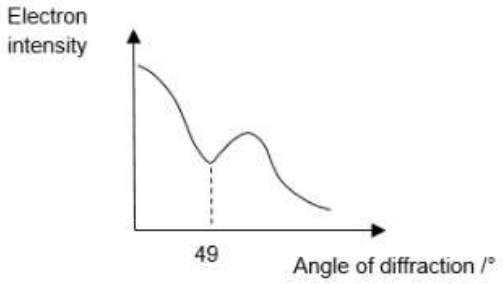
Question 44 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 10]

1.
A beam of electrons is incident on an iron foil ($Z = 56$) and electrons are diffracted. In the experiment, electrons have initial kinetic energy of $42\,GeV$. The first diffraction minimum is observed at an angle of 49° from the average position of most of the electrons detected.



1.
Calculate the density of the iron nucleus. [3]

2.
Using the diffraction information given, estimate the radius r of the iron nucleus . [4]

2.
In another experiment, alpha particles are emitted towards iron foil with an initial kinetic energy of $125\,keV$. Estimate the distance of the closest approach to the iron nucleus of the alpha particles. [3]

Question 45 Calculator Allowed

Markscheme

Difficulty: **Medium**

A pure radioactive sample of mass m has a decay constant of λ. Its activity is given by A. Another pure sample of the same material has a mass $2m$. Consider the following statements about the sample of mass $2m$ after an interval of two half lives.

- I. The decay constant is $\frac{\lambda}{2}$.
  II. The activity is $\frac{A}{2}$.
  III. The mass of decayed nucleons in the sample is $\frac{m}{2}$.

Which of the statements is/are correct?

- A. I only
- B. II only
- C. III only
- D. II and III only

Question 46 Calculator Allowed

Markscheme

Difficulty: **Medium**

The binding energy per nucleon of a ${}^{A}_{Z}\mathrm{X}$ nucleus is E. What is the mass defect for the formation of X from its constituent nucleons?

- A. $\frac{AE}{c^2}$
- B. $\frac{ZE}{c^2}$
- C. $\frac{E}{Ac^2}$
- D. $\frac{E}{Zc^2}$

Question 47 Calculator Allowed

Markscheme

Difficulty: **Medium**

1.
The following reaction takes place in a nuclear reactor:

$$^{235}_{92}\mathrm{U} + {}^{1}_{0}n \rightarrow {}^{144}_{56}\mathrm{Ba} + {}^{89}_{36}\mathrm{Kr} + 3{}^{1}_{0}n$$

1.
Differentiate between fission and fusion reactions. [3]
2.
Determine whether this reaction is a nuclear fission or nuclear fusion. [1]

2.
The following data is given for this reaction:
The binding energy per nucleon of $^{235}\mathrm{U}$ = 7.59 MeV
The binding energy per nucleon of $^{144}\mathrm{Ba}$ = 8.27 MeV
The binding energy per nucleon of $^{89}\mathrm{Kr}$ = 8.62 MeV
Calculate the energy released from this reaction. [3]

Question 48 Calculator Allowed

Markscheme

Difficulty: **Medium**

A nucleus $^{A}\mathrm{X}$ has a mass defect of N unified atomic mass units, Nu. What is the correct expression for its binding energy per nucleon?

- A. $931.5\frac{N}{A}\ MeV$
- B. $931.5\frac{N}{A}\ MeV\ c^{-2}$
- C. $931.5\frac{A}{N}\ MeV$
- D. $931.5\frac{N}{A}\ MeV\ c^{-2}$

Question 49 Calculator Allowed

Markscheme

Difficulty: **Medium**

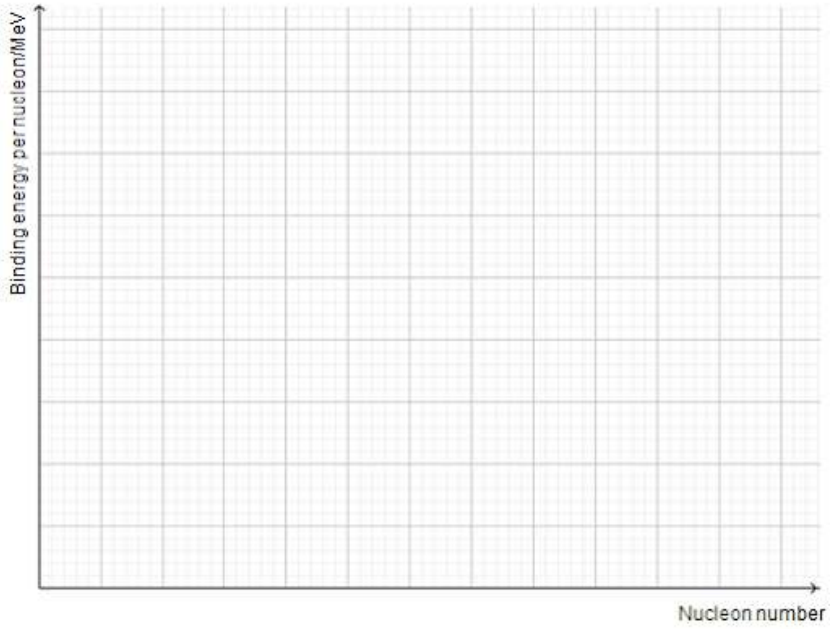
[Maximum Mark: 10]

1.
Uranium-235 ($^{235}_{92}U$) has a nuclear mass of 235.044 u.

1.
State what is meant by mass defect. [1]
2.
Calculate the mass defect of uranium-235. [2]

2.

1.
Calculate the binding energy per nucleon of uranium-235. [2]
2.
On the axes given, sketch the variation of the binding energy per nucleon with the nucleon number. Label the position of uranium-235. There is no need to add numbers to the axes. [3]



[Source: Created with GeoGebra (https://www.geogebra.org/graphing?lang=en)]

3.
Uranium-235 ($^{235}_{92}U$) decays into thorium-231 ($^{231}_{90}Th$).

- Determine which nuclide has a higher binding energy per nucleon. Use the graph to justify your answer. [2]

Question 50 Calculator Allowed

Markscheme

Difficulty: **Medium**

Which of the following is the most suitable method for managing nuclear waste?

- A. Bury them in the soil at shallow depths
- B. Dispose them into the ocean
- C. Store them in a disused mine
- D. Burn them in controlled incinerators

Question 51 Calculator Allowed

Markscheme

Difficulty: **Medium**

In the core of a star, a proton-proton chain reaction takes place.

- $4\ {}^{1}_{1}\mathrm{H} \rightarrow {}^{4}_{2}\mathrm{He} + 2\,{}^{0}_{+1}\beta^{+} + 2\,\nu_e + 2\,\gamma$

After the initial stage of the reaction, two positrons annihilate with electrons present in the surroundings. Which of the following correctly shows the nature of the chain reaction and the energy released due to annihilation?

	Nature of the chain reaction	Energy released
A.	Fusion	$(2)\,(0.511)\,MeV$
B.	Fusion	$(4)\,(0.511)\,MeV$
C.	Fission	$(2)\,(0.511)\,MeV$
D.	Fission	$(4)\,(0.511)\,MeV$

Question 52 Calculator Allowed

Markscheme

Difficulty: **Medium**

The parallax angles of Star A and Star B are 0.250 arcseconds and 0.500 arcseconds respectively. What is the shortest possible distance, in parsec, between Star A and Star B?

- A. $1\,pc$
- B. $2\,pc$
- C. $4\,pc$
- D. $6\,pc$

Question 53 Calculator Allowed

Markscheme

Difficulty: **Medium**

[Maximum mark: 12]

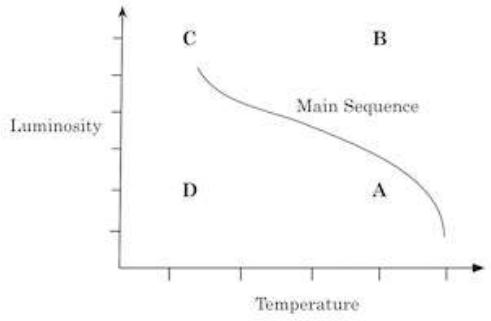
1. Define decay constant. [1]
2. Mushrooms contain a radioactive element called caesium-137, $^{137}_{55}\text{Cs}$. The element has a half-life of $30\,yr$ and a sample of living mushrooms has a constant decay rate of $2000\,Bq$ for every kg of mass. A sample of $50\,g$ is extracted from a dried mushroom and the activity is measured as $0.80\,Bq$.
    1. Caesium-137 decays to barium-137, $^{137}_{56}\text{Ba}$, by beta emission. Complete the nuclear reaction equation of this decay, showing both particles emitted in the decay.
    $^{137}_{55}\text{Cs} \rightarrow$ _____ + $^{137}_{56}\text{Ba}$ + _____ [2]
    2. Show that the decay constant of caesium-137 is about $0.023\,yr^{-1}$ [2]
    3. Given that the amount of caesium in the mushroom stops being replenished once the mushroom is no longer living, determine the age of the dried mushroom. [3]
3. Some other radioactive substances have half-lives of thousands of years. Outline the method used for determining the half-life of these kinds of substances. [4]

Question 54 Calculator Allowed

Markscheme

Difficulty: **Medium**

A star with a mass similar to the Sun, a temperature hotter than the Sun, and a radius similar to the Earth would be found in which region of the usual Hertzsprung-Russell diagram?



Question 55 Calculator Allowed

Markscheme

[Maximum marks: 8]

Difficulty: **Medium**

1.
A thorium-234 ($\mathrm{Th\text{-}234}$) nucleus undergoes beta decay and transforms into a protactinium-234 nucleus ($\mathrm{Pa\text{-}234}$). The binding energies per nucleon for Thorium-234 and protactinium-234 are $7.84\ \ MeV$ and $7.92\ \ MeV$ respectively. An electron has a rest mass energy of $0.020\ MeV$.
The reaction can be represented as follows:
$^{234}_{90}\mathrm{Th} \rightarrow\ ^{234}_{91}\mathrm{Pa} + 10e^{-} + \overline{\nu_e}$

1.
Explain what is meant by binding energy per nucleon. [2]

2.
Calculate, in joules, the total energy released in the decay. [3]

2.
A cubic container is filled with pure $\mathrm{thorium\text{-}234}$. The container does not contain any other elements. After 62 days, the ratio of the number of nuclei of decay product to $\mathrm{thorium\text{-}234}$ nuclei is 5.
Calculate, in days, the half-life of the $\mathrm{thorium\text{-}234}$ nuclei. [3]

Question 56 Calculator Allowed

Markscheme

Difficulty: **Medium**

The photocurrents produced by illuminating a metal plate with yellow light beams of intensities I_1 and I_2 are i_1 and i_2 respectively. The corresponding stopping voltages are V_1 and V_2. Given that $I_1 < I_2$, which of the following statements is true about the photocurrents and the stopping voltages?

	Photocurrents	**Stopping voltages**
A.	$i_1 < i_2$	$V_1 = V_2$
B.	$i_1 < i_2$	$V_1 < V_2$
C.	$i_1 > i_2$	$V_1 = V_2$
D.	$i_1 = i_2$	$V_1 < V_2$

Question 57 Calculator Allowed

Markscheme

Difficulty: **Medium**

A stationary nucleus of uranium-238 undergoes alpha decay to form thorium-234. The initial speed of the alpha particle is v. What is correct about the magnitude of the thorium-234 velocity and its direction?

	Magnitude	**Direction**
A.	$\frac{4}{234}v$	Same as the alpha particle
B.	$\frac{234}{238}v$	Opposite to the alpha particle
C.	$\frac{234}{238}v$	Same as the alpha particle
D.	$\frac{4}{234}v$	Opposite to the alpha particle

Question 58 Calculator Allowed

Markscheme

Difficulty: **Medium**

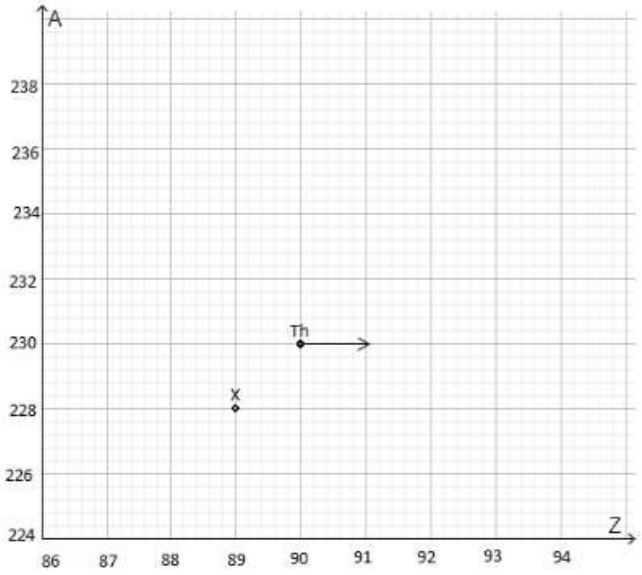
[Maximum mark: 10]
The plutonium isotope $^{238}_{94}\mathrm{Pu}$ undergoes alpha decay.

$^{238}_{\square}Pu \rightarrow\ ^{\square}_{92}U + ^{\square}_{\square}\square$

1.
Define isotope. [2]

2.
Rewrite this decay equation, filling in any missing information. [3]

3.
Plutonium undergoes a decay series. The graph below shows a part of this decay series.



[Source: Created with GeoGebra, - https://www.geogebra.org/graphing?lang=en]

1.
One type of decay cannot be represented on this diagram. Outline this type of decay and justify your answer. [2]

2.
State the type of decay of $^{230}_{90}\mathrm{Th}$ that is represented on the diagram above. Justify your answer. [2]

3.
Isotope X undergoes an alpha decay followed by a beta minus decay to produce isotope Y. Label the position of isotope Y on the diagram. [1]

4.
Compare the penetrating powers of alpha, beta and gamma radiation. [1]